Computer Programming Fundementals

Week 2

TEAM B

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What is computational thinking?

Systematic approach to problem solving

Approaching a complex problem in an organised and logical way in order to solve it.

Understanding and formulating problems

Identifying what the problem is, what needs to be solved, and what information is relevant.

Developing and describing solutions

Creating a logical solution and explaining it clearly, often in a way that could be followed by a computer.

The four cornerstones of computational thinking?

Decomposition

Breaking a complex problem into smaller, more manageable parts that can be solved individually.

Pattern recognition

Identifying similarities, patterns, or relationships between different problems or pieces of information.

Abstraction

Focusing on the most important information while ignoring unnecessary details that are not relevant to the problem.

Algorithmic thinking

Developing a clear and logical sequence of steps that can be followed to solve a problem or complete a task.

How to apply computational thinking to problem solving?

- **Decomposition: T**he problem can be decomposed into smaller parts so that we can deal with each part individually to identify what we need to focus onwhen solving this problem.

- **Pattern recognition:** Patterns can be identified to identify similarities and avoid solving the same type of problem from scratch in order to save time.

- **Abstraction:** It can be used to focus on important information, help plan a process, and remove unnecessary details. This allows you to break things down and understand how a process should be carried out, such as laying tasks out on a timeline. By reducing distractions and focusing on the information that actually matters, you can plan how to solve a problem more effectively.

- **Algorithms**: It can be used to create a clear sequence of steps to solve the problem efficiently from the information gathered from abstraction. this in order to create a clear and organised solution that can be followed and repeated.

Breaking down a Birthday Party Event

Today’s is Bob’s Birthday
Lets Plan His Event!



Using pattern Recognisition to solve sequences

a. 2, 4, 6, 8, ...

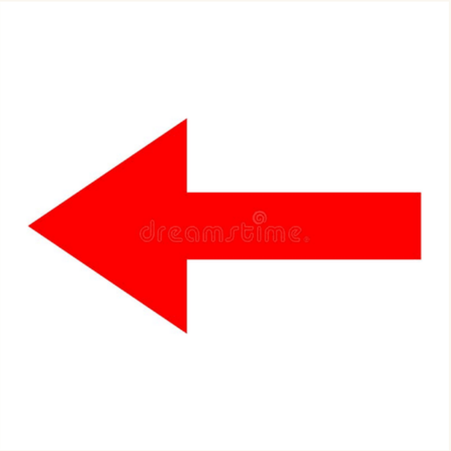
b. 5, 10, 20, 40,...

c. 1, 4, 9, 16, ...

How do we find a pattern from this?

Directions for a Friend



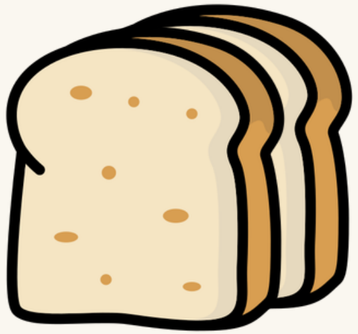


Your friend probably

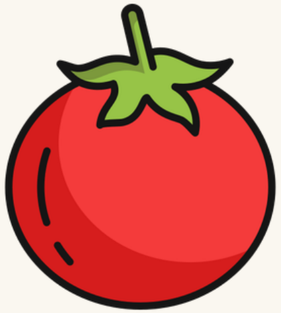
We can help him by using **abstraction...**

Making a Sandwich

Lets make a BLT sandwich with my recipie











Organising a Charity Event



The End

ANY QUESTIONS?